

REGAL Constraint Fitter -- Setup

HARD CONSTRAINTS (from public disclosures)

Trial start (first patient dosed) : Feb 8, 2021 (t = 0)
Enrollment completed : April 2024 (~38 months)
Total randomized : 126 (1:1, 63 per arm)

60 events triggered IA at month : 46 (~Dec 2024)
72 events update at month : 58 (~Dec 2025)
Final analysis trigger : 80 events

Median follow-up at IA : 13.5 +/- 2.0m (public disclosure)
Pooled (blinded) mOS at IA : > 12 months (public disclosure)
80-event status as of t=64 (May'26): t80 NOT in [63, 64] (no PR yet)
IDMC verdict at IA : Passed futility (continued)

HR success threshold (per SAP) : 0.636
Futility upper bound : HR_IA < 0.83
Efficacy lower bound : HR_IA > 0.4

ABC TOLERANCES

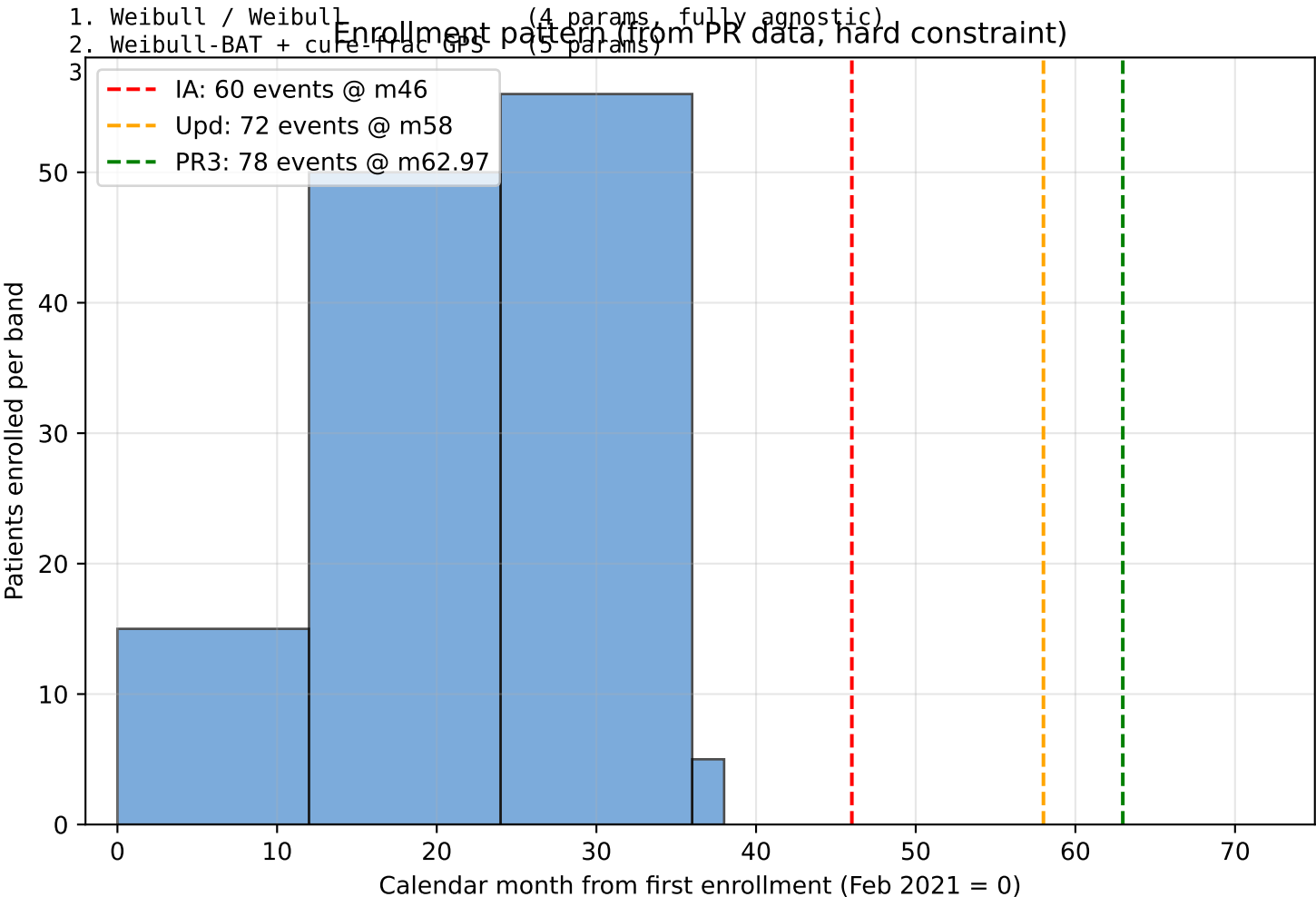
Event count IA : +/- 4
Event count Upd : +/- 4
Event count PR3 : +/- 2 (78 events at m62.97)
Increment IA->Upd : +/- 3 (observed: 12)
Increment Upd->PR3 : +/- 2 (observed: 6)

STRATIFIED OUTPUT

BAT mOS bin width : 1.0m

MODEL FAMILIES FIT

- 1. Weibull / Weibull (4 params, fully agnostic)
- 2. Weibull-BAT + cure-fraction (5 params)



Summary across all families

GLOBAL POSTERIORs (acceptance-weighted, marginalized over BAT mOS)
(See stratified pages for results conditional on each BAT mOS bin.)

=== Leaky-cure GPS ===

Compos accepted (sim phase): 1,331

BAT mOS (acc-weighted) : P05= 13.0m P50= 18.0m P95= 24.0m

HR @ 80 events : P05= 0.39 P50= 0.55 P95= 0.75

HEADLINE NUMBERS (acceptance-weighted means):

P(reach 80 events) : 100.0%

P(HR<0.636 | reached 80) : 71.4% <- CONDITIONAL

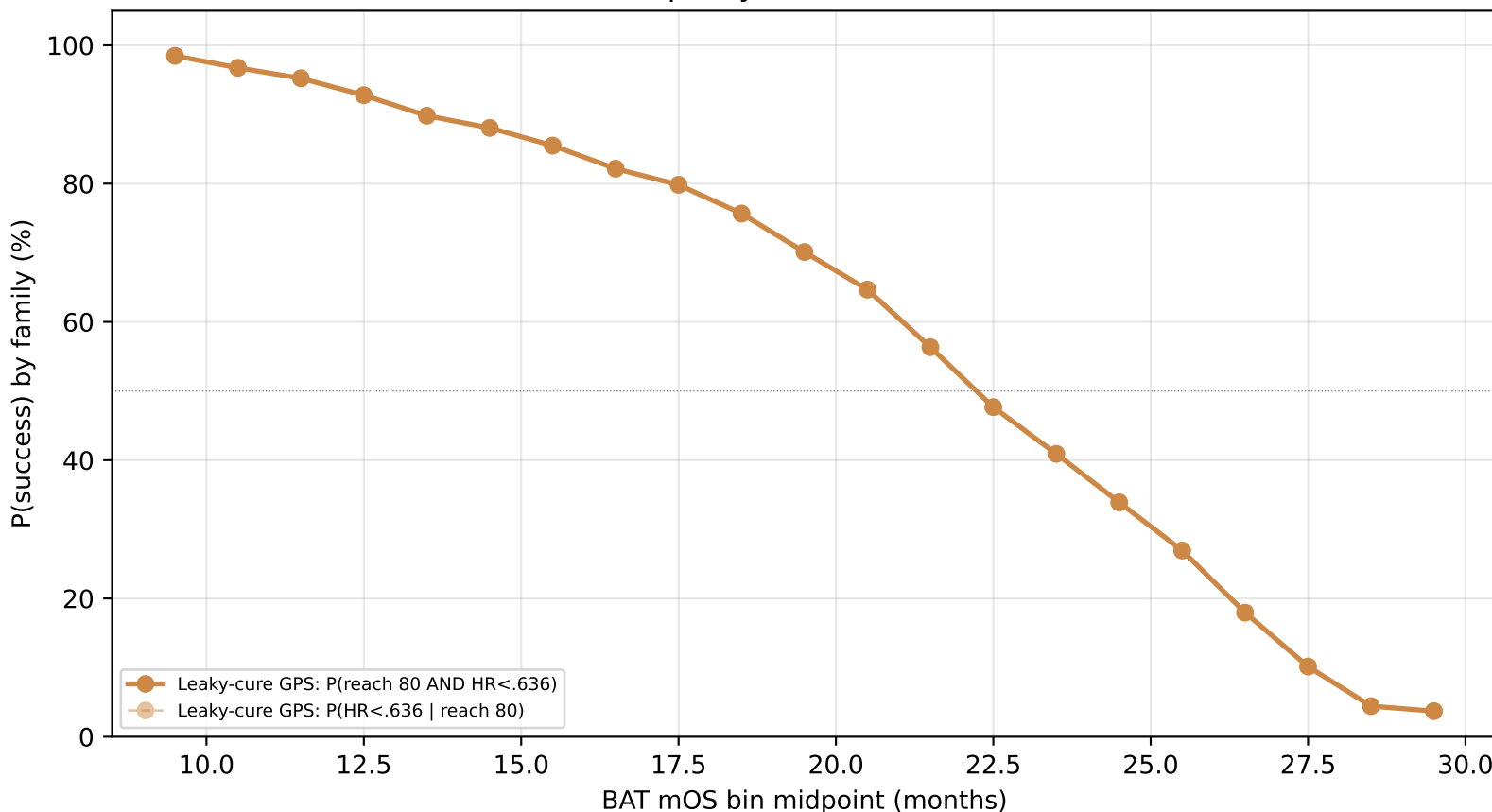
P(reach 80 AND HR<0.636) : 71.4% <- UNCONDITIONAL

P(no readout, never reach 80) : 0.0%

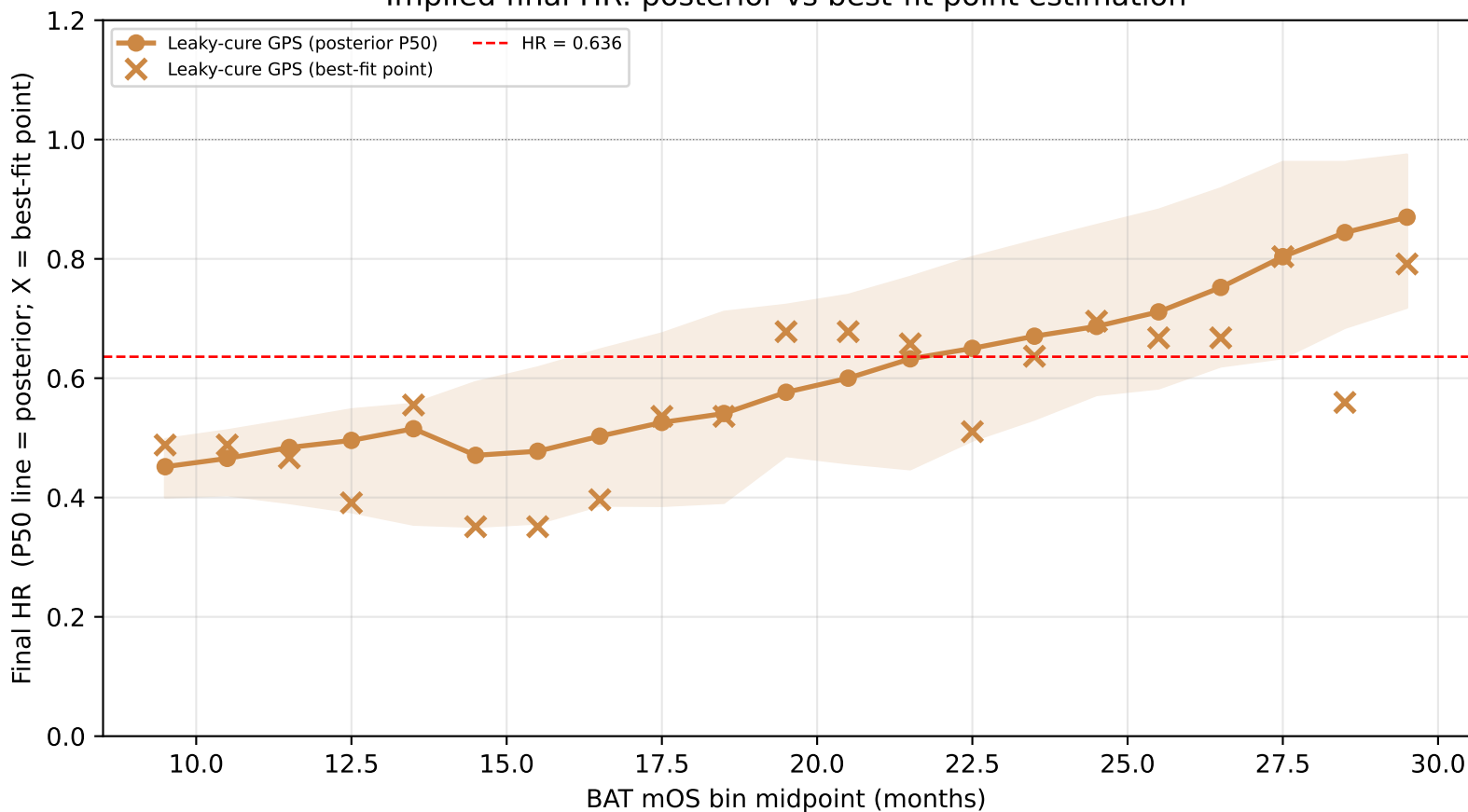
Calendar month of 80th : P05= 65.0m P50= 65.8m P95= 66.8m

Stratified by BAT mOS: P(success) and median HR per BAT bin

Stratified P(success): pick your BAT mOS, read off the answer



Implied final HR: posterior vs best-fit point estimation



Stratified posterior: Leaky-cure GPS

=== Leaky-cure GPS: P(success) and HR by BAT m0S ===

BAT m0S	n_combos	wt_share	P(success)	P(HR<.636)	P(reach80)	HR P50	HR P05	HR P95	t80 P50
9-10	26	0.2%	98.5%	98.5%	100.0%	0.452	0.400	0.498	66.0
10-11	49	0.6%	96.7%	96.7%	100.0%	0.466	0.404	0.512	65.5
11-12	84	1.5%	95.2%	95.2%	100.0%	0.484	0.391	0.530	65.7
12-13	110	2.6%	92.8%	92.8%	100.0%	0.496	0.376	0.548	65.8
13-14	125	4.3%	89.8%	89.8%	100.0%	0.515	0.355	0.557	65.7
14-15	142	7.0%	88.1%	88.1%	100.0%	0.471	0.351	0.593	65.5
15-16	146	9.3%	85.5%	85.5%	100.0%	0.478	0.357	0.618	65.6
16-17	143	10.6%	82.2%	82.2%	100.0%	0.503	0.387	0.648	65.7
17-18	156	11.9%	79.8%	79.8%	100.0%	0.526	0.386	0.675	65.8
18-19	166	12.0%	75.7%	75.7%	100.0%	0.541	0.391	0.711	65.9
19-20	157	10.0%	70.1%	70.1%	100.0%	0.576	0.470	0.723	65.9
20-21	155	8.1%	64.7%	64.7%	100.0%	0.600	0.457	0.740	65.9
21-22	169	6.6%	56.3%	56.3%	100.0%	0.632	0.447	0.770	65.9
22-23	192	5.3%	47.7%	47.7%	100.0%	0.650	0.496	0.803	66.0
23-24	206	4.1%	40.9%	40.9%	100.0%	0.670	0.531	0.830	65.9
24-25	185	2.6%	33.9%	33.9%	100.0%	0.687	0.572	0.857	65.8
25-26	148	1.4%	26.9%	26.9%	100.0%	0.711	0.583	0.882	65.9
26-27	118	0.8%	17.9%	17.9%	100.0%	0.752	0.620	0.918	65.9
27-28	78	0.5%	10.2%	10.2%	100.0%	0.804	0.634	0.962	65.6
28-29	48	0.3%	4.4%	4.4%	100.0%	0.844	0.685	0.962	65.6
29-30	36	0.2%	3.7%	3.7%	100.0%	0.870	0.718	0.975	65.7

P(success) = P(reach 80 AND HR < 0.636)

P(HR<.636) = conditional on reaching 80

HR P50/05/95= median of per-combo median_HR_final, weighted by combo acceptance rate

t80 P50 = posterior median calendar month of 80th event

Best-fit point estimate: Leaky-cure GPS

=== Leaky-cure GPS: BEST-FIT POINT per BAT mOS bin ===

(residual-minimization, max abs |E[ev_IA]-60|, |E[ev_upd]-72|, |t80-64|)

BAT_bin	bat	bat_sh	cure	unc_med	unc_sh	leak	resid	HR_fin	P_succ	t80
9-10	10.000	0.600	0.600	24.000	1.000	0.090	0.61	0.488	100.0%	64.4
10-11	10.000	0.600	0.600	24.000	1.000	0.090	0.61	0.488	100.0%	64.4
11-12	11.000	0.800	0.500	28.000	1.000	0.040	0.89	0.467	100.0%	63.4
12-13	12.000	0.800	0.800	6.000	0.500	0.100	0.93	0.391	100.0%	64.6
13-14	14.000	0.600	0.300	22.000	1.000	0.000	0.96	0.555	72.3%	65.1
14-15	15.000	1.000	0.800	4.000	2.000	0.080	0.82	0.351	100.0%	64.9
15-16	15.000	1.000	0.800	4.000	2.000	0.080	0.82	0.351	100.0%	64.9
16-17	16.000	1.000	0.700	4.000	0.500	0.070	1.12	0.396	98.3%	65.2
17-18	18.000	0.800	0.300	24.000	0.500	0.100	0.83	0.536	77.4%	64.9
18-19	18.000	0.800	0.300	24.000	0.500	0.100	0.83	0.536	77.4%	64.9
19-20	20.000	0.600	0.300	18.000	1.000	0.040	0.97	0.678	33.3%	64.5
20-21	20.000	0.600	0.300	18.000	1.000	0.040	0.97	0.678	33.3%	64.5
21-22	22.000	0.800	0.100	24.000	0.500	0.090	0.87	0.658	42.3%	64.9
22-23	23.000	1.000	0.600	4.000	1.500	0.070	0.66	0.510	88.9%	64.7
23-24	24.000	0.800	0.300	12.000	0.500	0.090	0.66	0.637	45.5%	64.6
24-25	25.000	0.800	0.100	18.000	0.500	0.030	0.64	0.696	25.0%	64.7
25-26	26.000	0.800	0.200	14.000	0.500	0.080	0.41	0.668	37.5%	64.2
26-27	26.000	0.800	0.200	14.000	0.500	0.080	0.41	0.668	37.5%	64.2
27-28	27.000	0.800	0.300	10.000	0.500	0.100	0.49	0.804	16.7%	64.2
28-29	28.000	0.800	0.000	20.000	0.500	0.040	0.69	0.559	100.0%	64.5
29-30	30.000	0.800	0.000	18.000	0.500	0.080	0.67	0.791	0.0%	64.2

resid = max abs residual to event anchors (events / months)

HR_fin = HR at 80 events for THIS single best-fit combo (not averaged)

P_succ = P(success) for THIS single combo's simulations

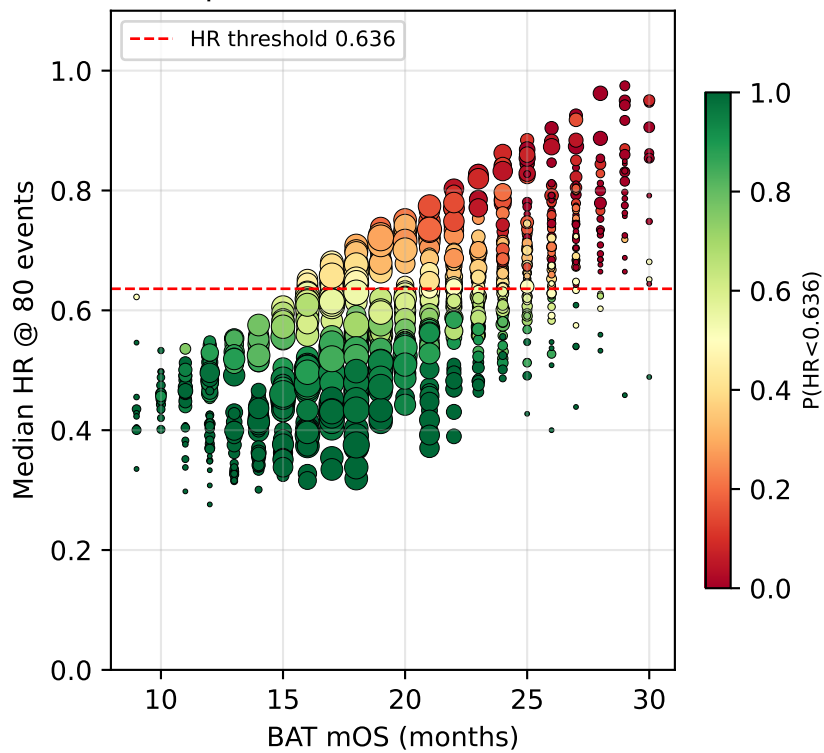
t80 = calendar month of 80th event for this combo

This is yg19-style point estimation for direct comparison.

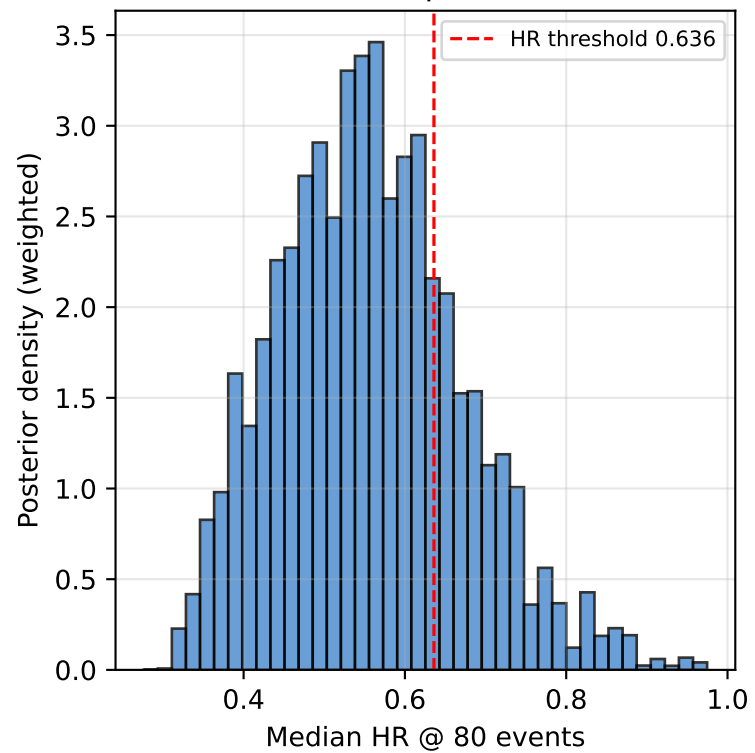
Use the stratified-posterior table for averaged-over-uncertainty answers.

Leaky-cure GPS (page 1/2)

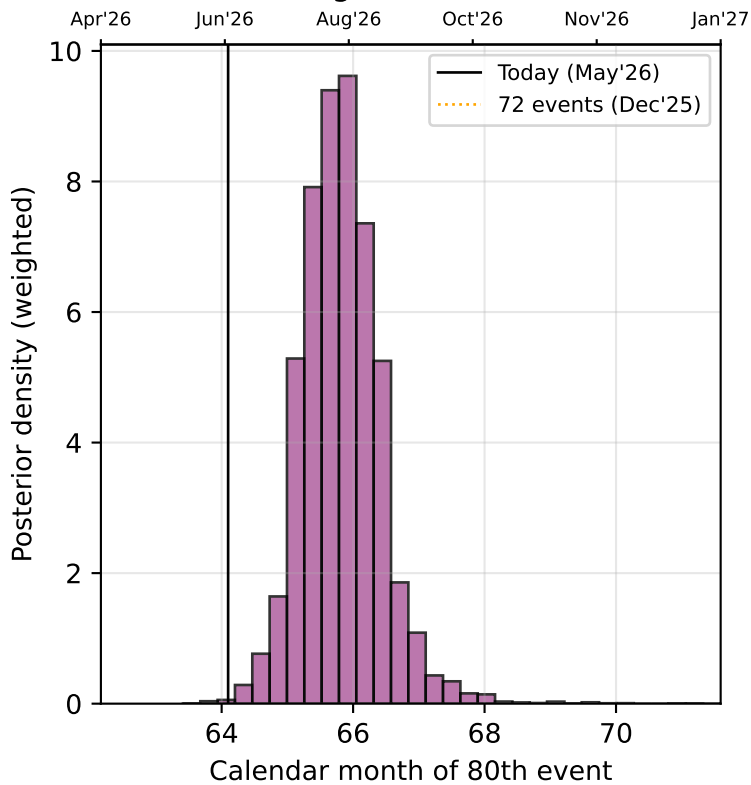
Implied final HR vs BAT mOS



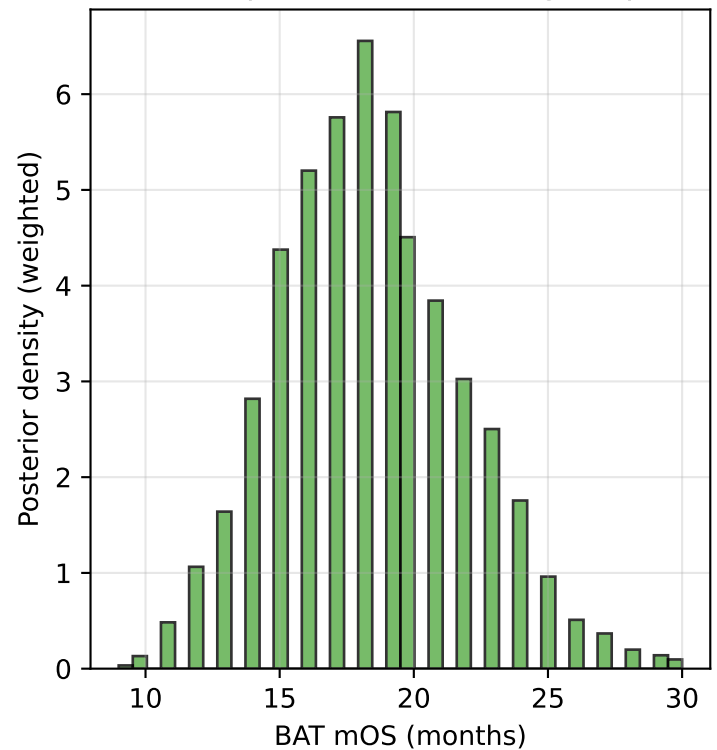
Final-HR posterior



Timing of 80th event

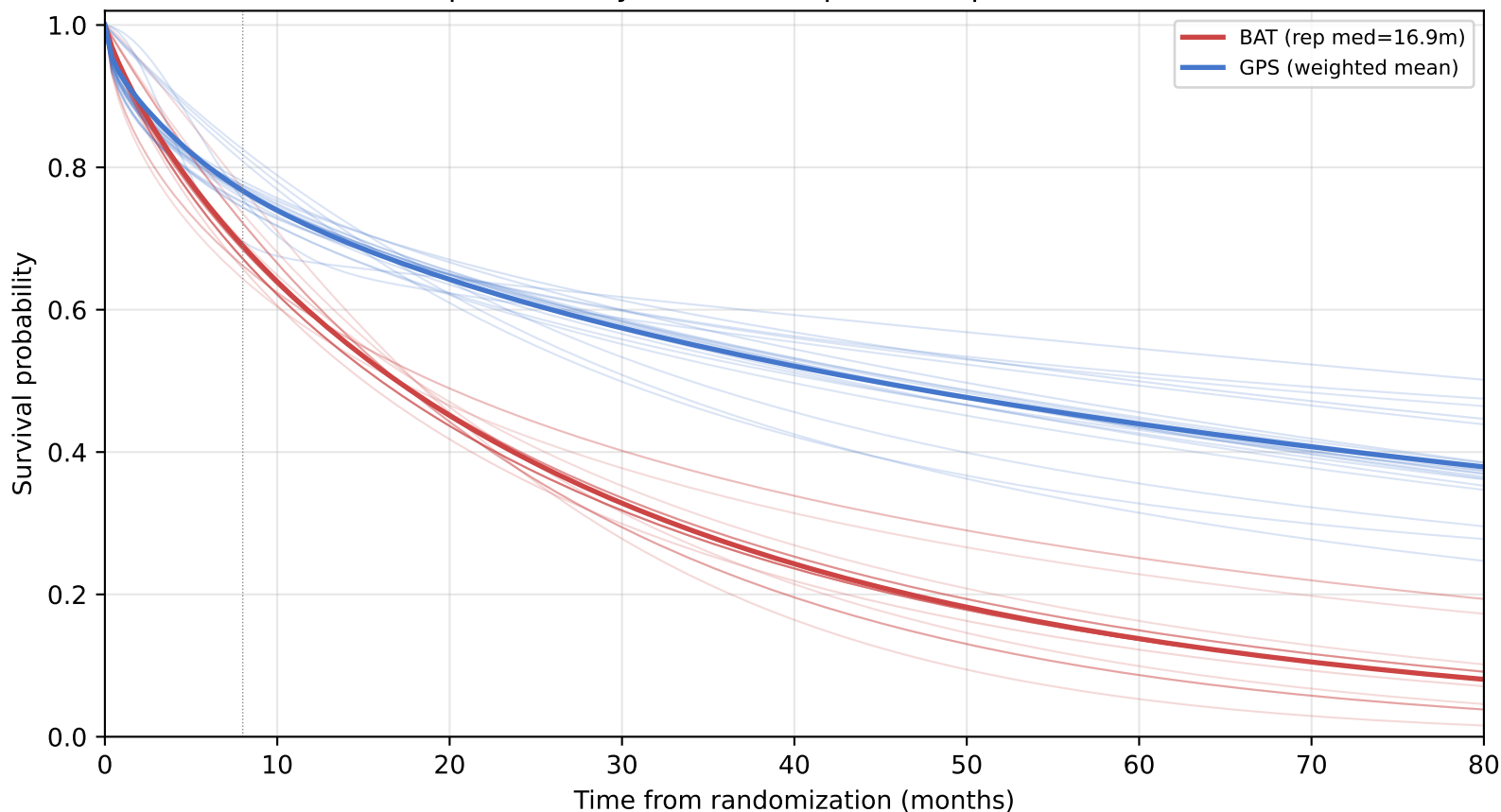


BAT mOS posterior (no biological prior)

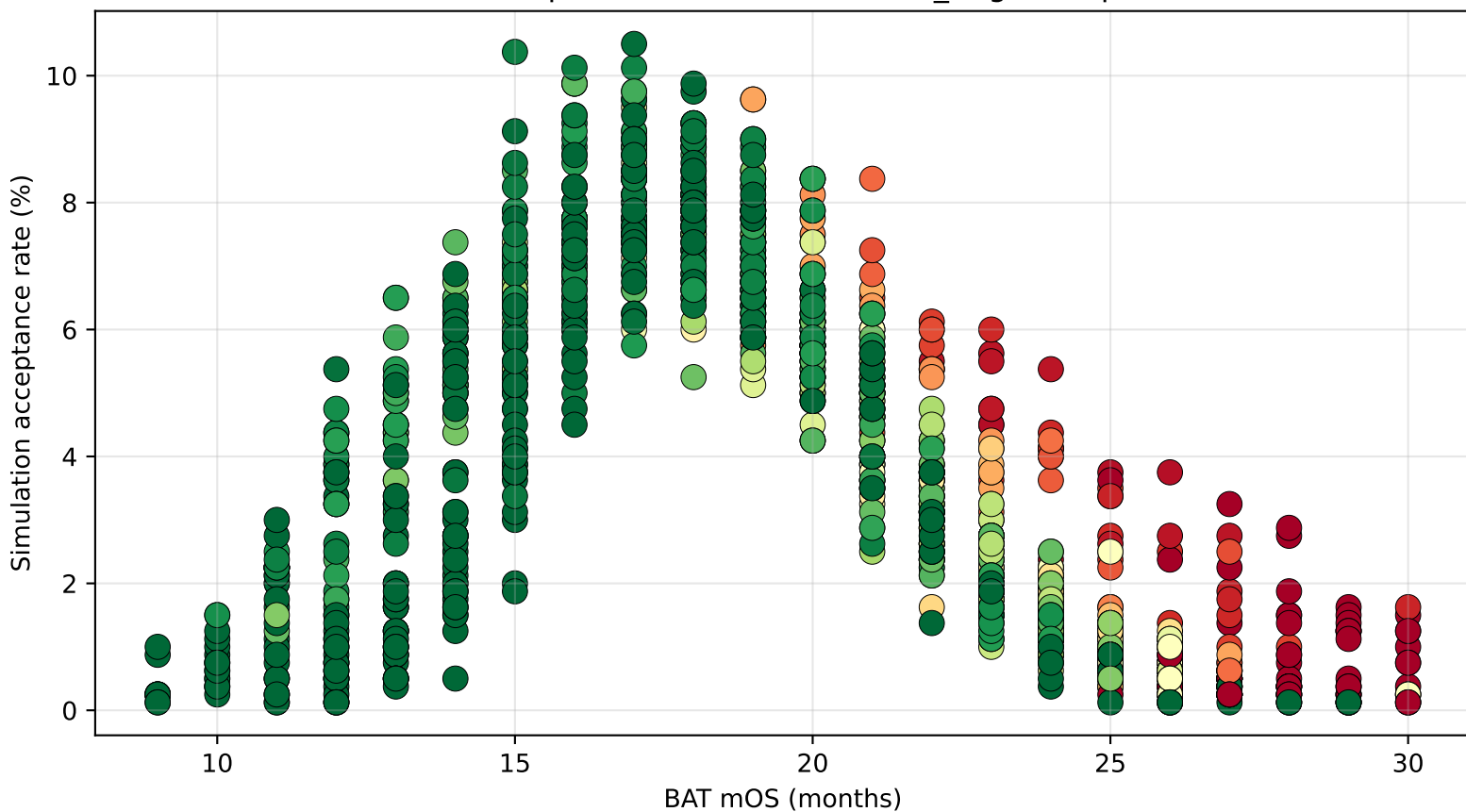


Leaky-cure GPS (page 2/2)

Implied KM-style curves (top 20 accepted combos)

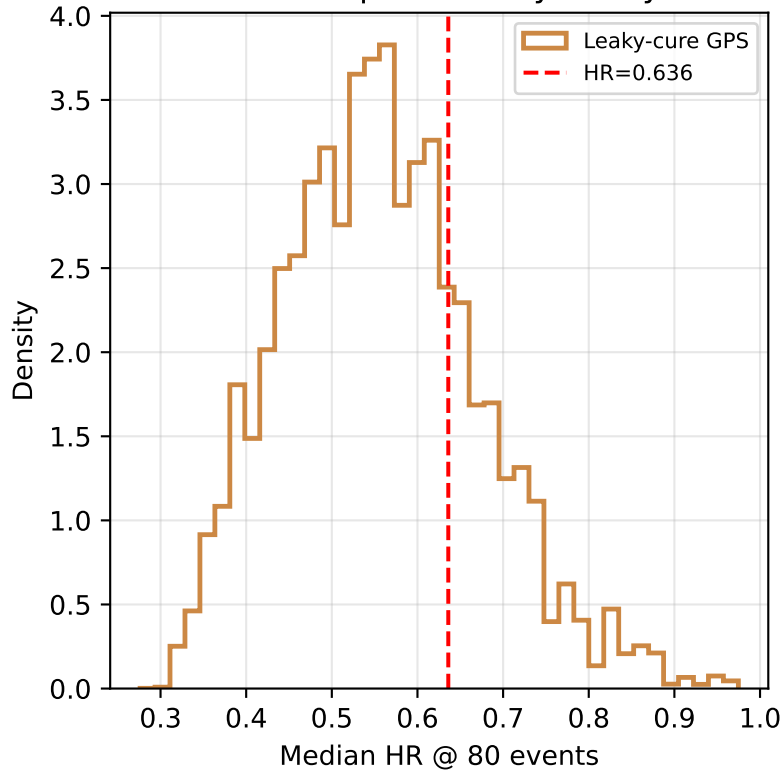


Simulation acceptance rate (events + HR_IA gate + pool mOS)

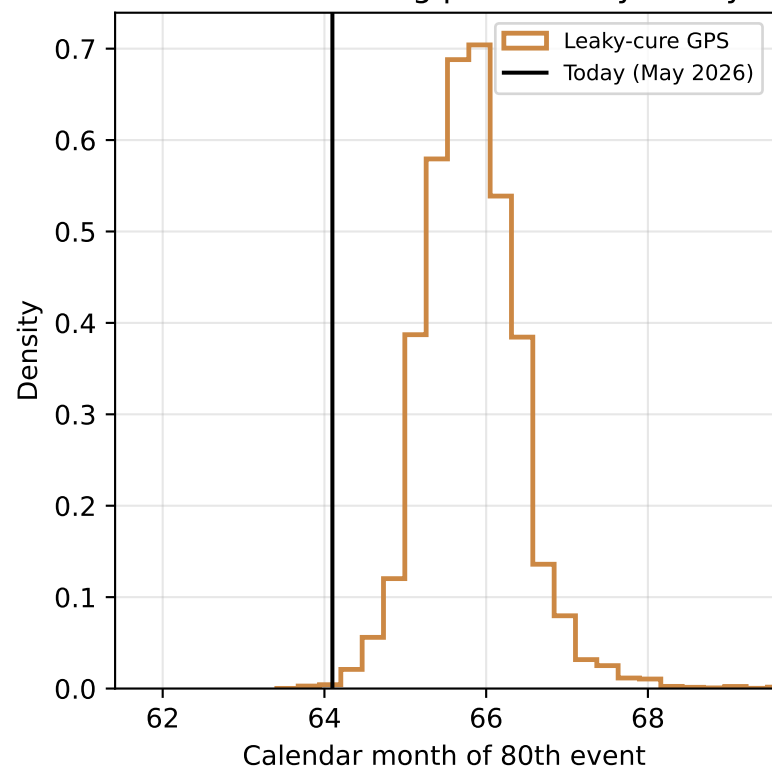


Cross-family comparison

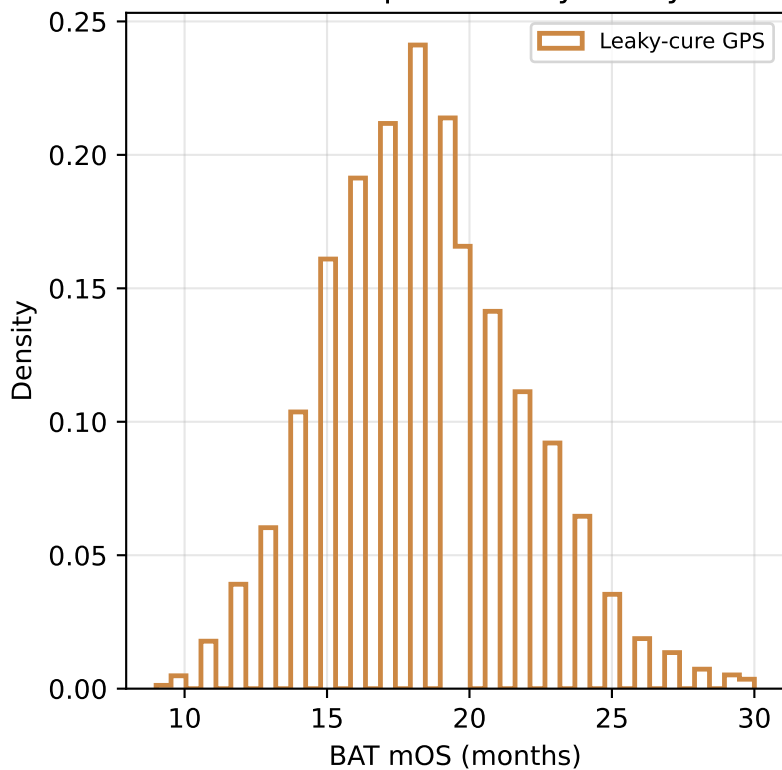
Final-HR posterior by family



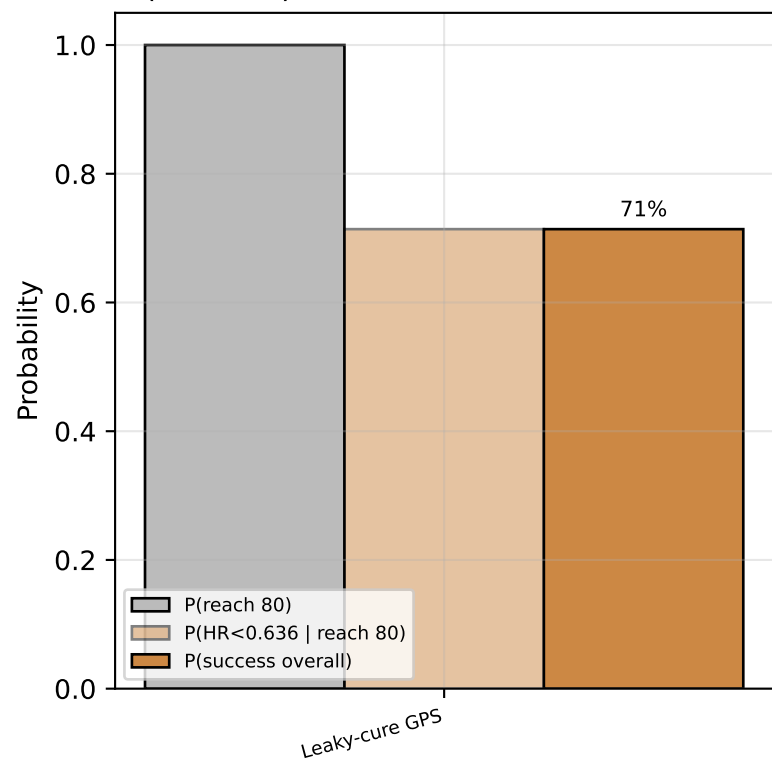
80th-event timing posterior by family



BAT mOS posterior by family



P(success): unconditional vs conditional



Top accepted combos

TOP-20 ACCEPTED COMBOS PER FAMILY (by acceptance rate)

--- Leaky-cure GPS ---

exp_ev_pr3	bat_med	bat_shape	cure_frac	unc_med	unc_shape	leak_yr	acceptance_rate
77.062	17.000	1.000	0.300	28.000	0.500	0.040	0.105
77.320	15.000	0.800	0.600	10.000	0.500	0.090	0.104
77.186	16.000	0.800	0.600	8.000	0.500	0.090	0.101
76.630	17.000	0.800	0.400	20.000	0.500	0.090	0.101
77.010	16.000	0.800	0.400	22.000	0.500	0.090	0.099
77.039	16.000	0.800	0.400	20.000	0.500	0.080	0.099
77.616	16.000	0.800	0.400	22.000	0.500	0.100	0.099
77.401	18.000	1.200	0.700	4.000	2.000	0.050	0.099
77.263	17.000	0.800	0.300	26.000	0.500	0.100	0.098
77.060	17.000	0.800	0.600	8.000	0.500	0.100	0.098
76.927	18.000	1.000	0.700	6.000	2.000	0.070	0.098
77.477	17.000	0.800	0.200	28.000	0.500	0.070	0.096
77.388	17.000	1.000	0.300	20.000	0.500	0.000	0.096
77.401	19.000	0.600	0.300	18.000	1.000	0.030	0.096
77.240	19.000	0.600	0.400	18.000	1.000	0.090	0.096
76.810	17.000	0.600	0.400	18.000	1.000	0.060	0.095
77.430	16.000	0.800	0.500	14.000	0.500	0.090	0.094
77.426	16.000	0.800	0.500	16.000	0.500	0.100	0.094
77.562	17.000	1.000	0.400	14.000	0.500	0.020	0.094
77.018	16.000	0.800	0.400	24.000	0.500	0.100	0.092